Vietnam National University of HCMC

International University

School of Computer Science and Engineering

Program of MIT

Course: Advanced Computer Graphics (IT516)

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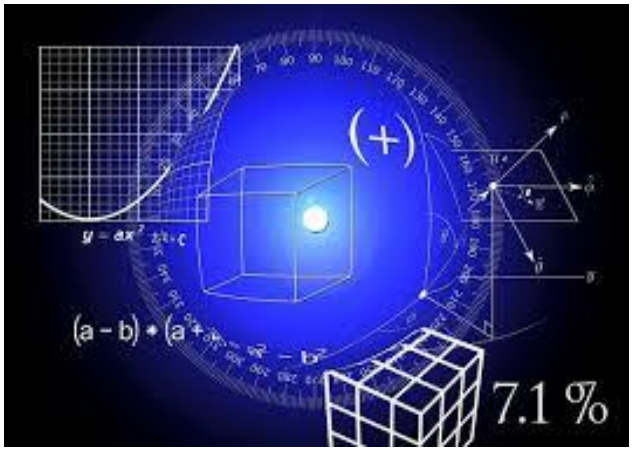
Mathematical Background

(The lecture in 13/09/2024)



1) Refer chapters 5, 7, 8, 9. John Vince, Mathematics for Computer Graphics, 5th Edition, Springer 2017
2) Refer chapters 1, 3, 4. John Vince, Vector Analysis for Computer Graphics, 2nd Edition, Springer 2021
3) Lecture note - Fundamentals of Computer Graphics 2019

Acknowledgement

- The taught material in this course draws upon a mathematical background in linear algebra.
- The mathematical foundation include numbers systems, algebra, trigonometry, 2D and 3D geometry, vectors, equations, matrices, determinants and calculus.
- The lectures have been summarized from the textbooks, lecture notes on field of computer graphics, mathematics for computer graphics all over the world.
- I would like to thank all Professors who created such a good work on those textbooks and lecture notes.
- I collect and modify them based on the textbooks to create this slide to review the background of mathematics for computer graphics.

Objectives

- Mathematics for CG
- Points, vectors and notation
- Basic Vectors Algebra
- Vector multiplication
- Reference frames
- Cartesian vs Radial-Polar Form
- Matrix Algebra

Notations ➔

α	alpha	ν	nu
β	beta	ξ	xi
γ	gamma	o	o
δ	delta	π	pi
ε	epsilon	ρ	rho
ζ	zeta	σ	sigma
η	eta	τ	tau
θ	theta	υ	upsilon
ι	iota	ϕ	phi
κ	kappa	χ	chi
λ	lambda	ψ	psi
μ	mu	ω	omega

and some upper-case symbols:

Γ	Gamma	Σ	Sigma
Δ	Delta	Υ	Upsilon
Θ	Theta	Φ	Phi
Λ	Lambda	Ψ	Psi
Ξ	Xi	Ω	Omega
Π	Pi		

Basic elements

- Geometry is the study of the relationships among objects in an n-dimensional space
  - In computer graphics, we are interesting the objects that exist in three dimensions
- Want a minimum set of primitives from which we can build more sophisticated objects
- We will need three basic elements
  - Scalars
  - Vectors
  - Points

Coordinate-free geometry

- When we learned simple geometry, most of us started with a Cartesian approach
  - Points were at locations in space **p** = (x, y, z)
  - We derived results by algebraic manipulations involving these coordinates
- This approach was nonphysical
  - Physically, points exist regardless of the location of an arbitrary coordinate system
  - Most geometric results are independent of the coordinate system
  - Euclidean geometry: two triangles are identical if two corresponding sides and the angle between them are identical

Geometry

- Geometry provides a mathematical foundation of objects description in computer graphics:
  - Geometric Spaces: Vector, Affine, Euclidean, Cartesian, Projective
  - Affine Geometry
  - Affine Transformations
  - Perspective
  - Projective Transformations
  - Matrix Representation of Transformations
  - Viewing Transformations

Introduction

- **Points** are associated with locations of space
- **Vectors** represent displacements between points or directions
- Points, vectors, and operators that combine them are the common tools for solving many geometric problems that arise in Geometric Modeling, Computer Graphics, Animation, Visualization, and Computational Geometry.

Introduction

- The three most fundamental operators on vectors are **the dot product, the cross product,** and **the mixed product** (sometimes called the triple product).
- Although a point and a vector may be represented by its three coordinates, they are of **different type** and should not be mixed
- **Avoid**, whenever possible, **using coordinates** when formulating geometric constructions. Instead use vectors and points.

What will you learn here?

- Properties of dot, cross, and mixed products
- How to write simple tests for
  - 4-points and 2-lines coplanarity
  - Intersection of two coplanar edges
  - Parallelism of two edges or of two lines
  - Clockwise orientation of a triangle from a viewpoint
  - Positive orientation of a tetrahedron
  - Edge/triangle and ray/triangle intersection

What will you learn here?

- How to compute
  - Center of mass of a triangle
  - Volume of a tetrahedron
  - “Shadow” (orthogonal projection) of a vector on a plane
  - Line/plane intersection
  - Plane/plane intersection
  - Plane/plane/plane intersection

Terminology

- Orthogonal to = Normal to = forms a 90° angle with
- Norm of a vector = length of the vector
- Are coplanar = there is a plane containing them

Some of notations

≡ means “is always equal to” or “by definition”

== means the test “is equal to” used in Boolean expressions

:= is the assignment “is computed as” and is used in construction algorithms

s (italic type) Boolean

v (normal type) scalar

a (bold type lower-case) point

A (bold upper-case) pointset (edge, curve, triangle, surface, solid)

U (bold upper-case) vector

u (bold lower-case) unit vector

T (bold upper-case) transformation

Scalars

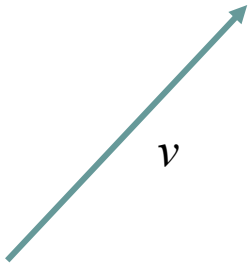
- Need three basic elements in geometry
  - Scalars, Vectors, Points
- Scalars can be defined as members of sets which can be combined by two operations (addition and multiplication) obeying some fundamental axioms (association, commutation, inverses)
- Examples include the real and complex number systems under the ordinary rules with which we are familiar
- Scalars alone have no geometric properties

Vectors and point

- We commonly use vectors to represent:
  - Points in space (i.e., location)
  - Displacements from point to point
  - Direction (i.e., orientation)
- But we want points and directions to behave differently
  - Ex: To *translate* something means to move it without changing its orientation
  - Translation of a point = different point
  - Translation of a direction = same direction

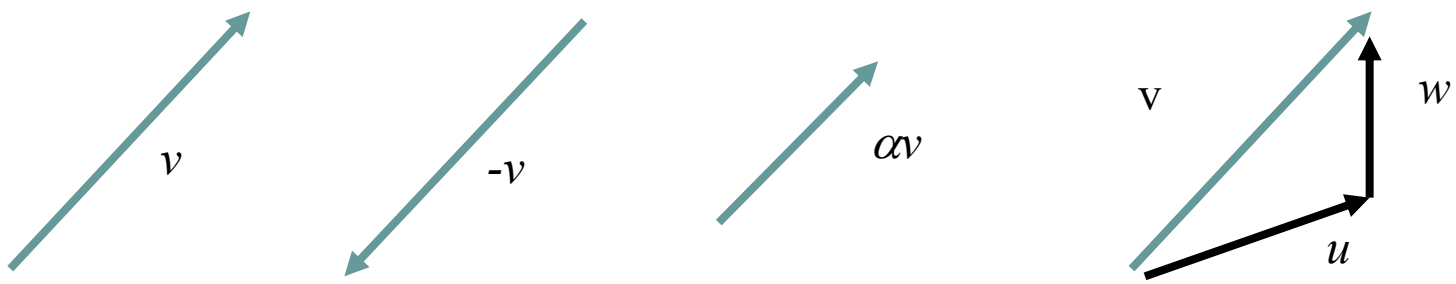
Vectors

- Physical definition: a vector is a quantity with two attributes
  - Direction
  - Magnitude
- Examples include
  - Force
  - Velocity
  - Directed line segments
    - Most important example for graphics
    - Can map to other types



Vector operations

- Every vector has an inverse
  - Same magnitude but points in opposite direction
- Every vector can be multiplied by a scalar
- There is a zero vector
  - Zero magnitude, undefined orientation
- The sum of any two vectors is a vector
  - Use head-to-tail axiom



Vectors

- **Vector space** (analytic definition from Descartes in the 1600s)
  - $\mathbf{U}+\mathbf{V}=\mathbf{V}+\mathbf{U}$ (vector addition)
  - $\mathbf{U}+(\mathbf{V}+\mathbf{W})=(\mathbf{U}+\mathbf{V})+\mathbf{W}$
  - $\mathbf{0}$ is the zero (or null) vector if $\forall\mathbf{V}$, $\mathbf{0}+\mathbf{V}=\mathbf{V}$
  - $-\mathbf{V}$ is the inverse of $\mathbf{V}$, so the vector subtraction $\mathbf{V}-\mathbf{V}=\mathbf{0}$
  - $s(\mathbf{U}+\mathbf{V})=s\mathbf{U}+s\mathbf{V}$ (scalar multiplication)
  - $\mathbf{U}/s$ is the scalar division (same as multiplication by s^{-1})
  - $(a+b)\mathbf{V}=a\mathbf{V}+b\mathbf{V}$

Vectors

- Vectors are used to represent displacement between points
- Each vector V has a norm (length) denoted ||V||
  - **V** / ||**V**|| is the unit vector (length 1) of **V**. We denote it $\mathbf{V_u}$
  - If ||**V**||==**0**, then **V** is the null vector **0**
- Unit vectors are used to represent
  - Basis vectors of a coordinate system
  - Directions of tangents or normals in definitions of lines or planes

Vector spaces

- A *linear combination* of vectors results in a new vector:

$$\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots + \alpha_n\mathbf{v}_n$$

- If the only set of scalars such that

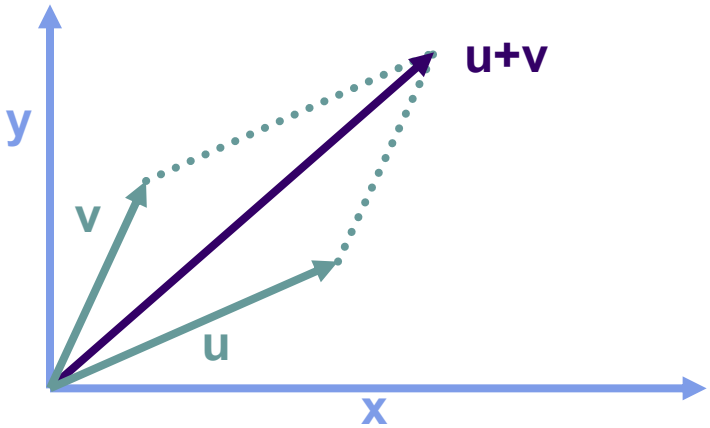
$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots + \alpha_n\mathbf{v}_n = \mathbf{0}$$

  is $\alpha_1 = \alpha_2 = \ldots = \alpha_3 = \mathbf{0}$

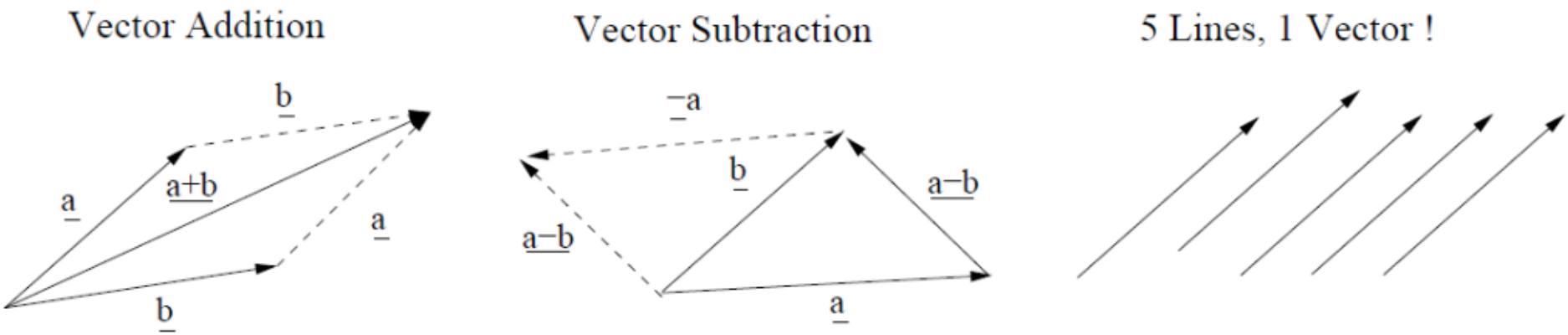
  then we say the vectors are *linearly independent*
- The *dimension* of a space is the greatest number of linearly independent vectors possible in a vector set
- For a vector space of dimension *n*, any set of *n* linearly independent vectors form a *basis*

Vector spaces: an example

- Our common notion of vectors in a 2D plane is (you guessed it) a vector space:
  - Vectors are “arrows” rooted at the origin
  - Scalar multiplication “streches” the arrow, changing its length (*magnitude*) but not its direction
  - Addition uses the “trapezoid rule”:



Examples of vector computation



Illustrating vector addition (left) and subtraction (middle). Right: Vectors have direction and magnitude; lines (sometimes called ‘rays’) are vectors plus a starting point.

Vector computation:

- Magnitude: abs(a) = sqrt($u^2 + v^2$)
- Normalization: Norm(a) = a/abs(a)
- Dot product: $a \circ b = a_1b_1 + a_2b_2 + a_3b_3 + \ldots + a_nb_n$; $a \circ b = |a||b|\cos\theta$
- Cross product:

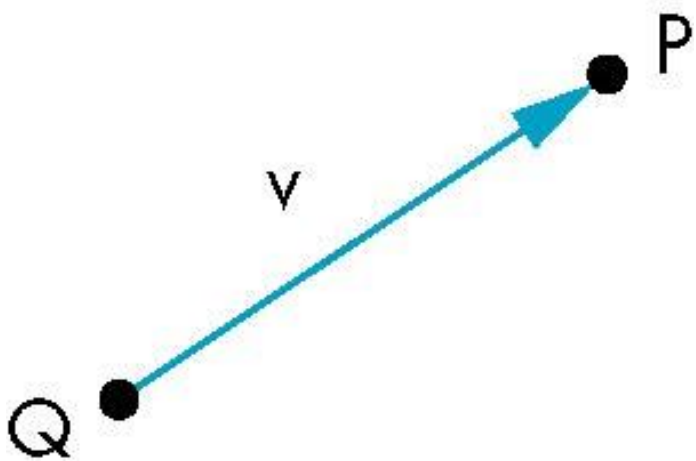
$$\underline{a} \times \underline{b} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}$$

Vector spaces: basis vectors

- Given a basis for a vector space:
  - Each vector in the space is a *unique* linear combination of the basis vectors
  - The *coordinates* of a vector are the scalars from this linear combination
  - Best-known example: Cartesian coordinates
    - An example: 3D coordinate (x, y, z) space
  - Note that a given vector $\mathbf{v}$ will have different coordinates for different bases

Points

- Location in space
- Operations allowed between points and vectors
  - Point-point subtraction yields a vector
  - Equivalent to point-vector addition



$$v = P - Q$$

$$P = v + Q$$

Affine space

Definition: Set of Vectors ***V*** and a Set of Points ***P***

- Vectors ***V*** form a vector space.
- Points can be combined with vectors to make new points:

$P + v = Q$ with $P, Q \in \boldsymbol{P}$ and $v \in \boldsymbol{V}$

Dimension: The dimension of an affine space is the same as that of **V** .

Affine space

Note:

- All other point operations are just variations on the

  P + $\underline{v}$ operation.
- No distinguished origin
- No notion of distances or angles.
- Most of what we do in graphics is affine.

Affine space

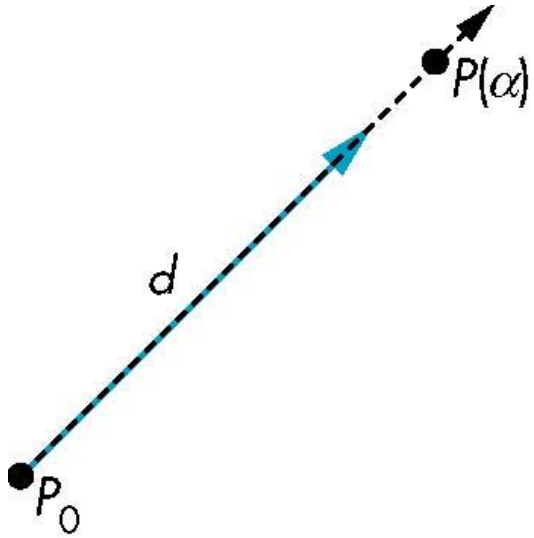
- Defined together with a vector space and the point difference mapping
  - **V = b - a** (vector **V** is the displacement from point a to point b)
  - Notation: ab stands for the vector **b – a**
  - **c - a** = **(c - b)** + **(b** – **a).** Written **ac** = **ab** + **bc**
  - Point b is uniquely defined, given point a and vector **(b - a)**

Affine space

- If we pick a as origin, then there is a 1 to 1 mapping between a point b and the vector **b - a**.
- We may extend the vector operations to points, but usually only when the result is independent of the choice of the origin!
  - **A** + **b** is not allowed (the result depends on the origin)
  - (**a** + **b**)/2 is OK
- Points may be used to represent
  - The vertices of a triangle or polyhedron
  - The origin of a coordinate system
  - Points in the definition of lines or plane

Lines

- Consider all points of the form
  - $P(\alpha)=P_0 + \alpha\, \mathbf{d}$
  - Set of all points that pass through P_0 in the direction of the vector $\mathbf{d}$



Parametric form

- This form is known as the parametric form of the line
  - More robust and general than other forms
  - Extends to curves and surfaces
- Two-dimensional forms
  - Explicit: $y = mx + h$
  - Implicit: $ax + by + c = 0$
  - Parametric:

$$x(\alpha) = \alpha x_0 + (1-\alpha)x_1$$

$$y(\alpha) = \alpha y_0 + (1-\alpha)y_1$$

Rays and line segments

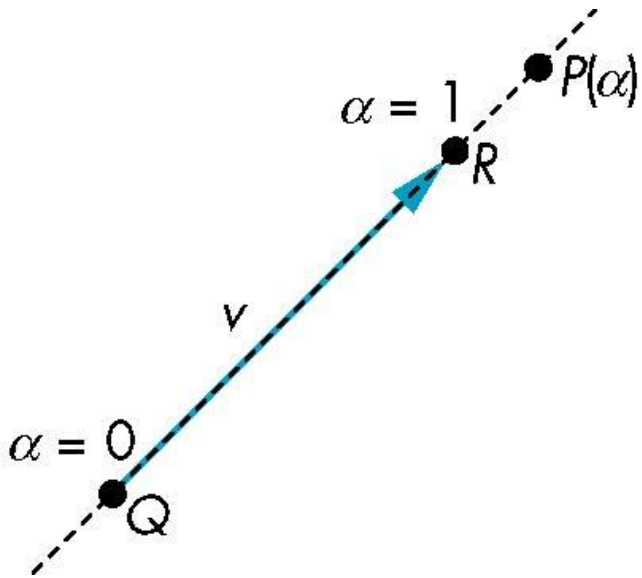
- If $\alpha >= 0$, then $P(\alpha)$ is the *ray* leaving P_0 in the direction $\mathbf{d}$

If we use two points to define v, then

$P(\alpha) = Q + \alpha (R-Q) = Q + \alpha v$

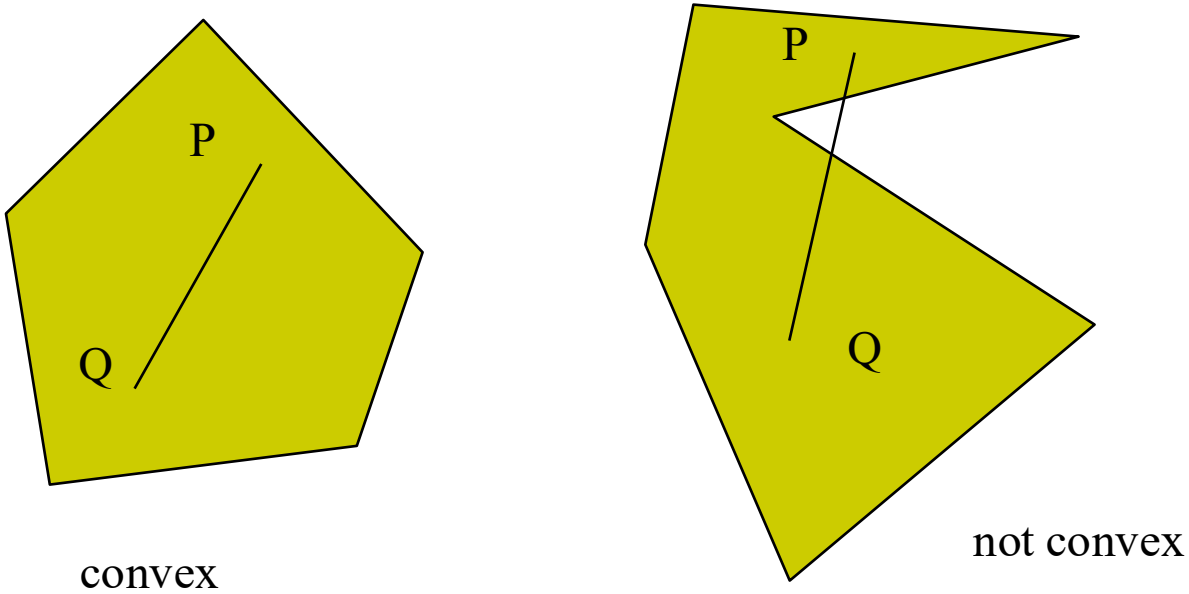
$= \alpha R + (1-\alpha) Q$

For $0 <= \alpha <= 1$ we get all the points on the *line segment* joining R and Q



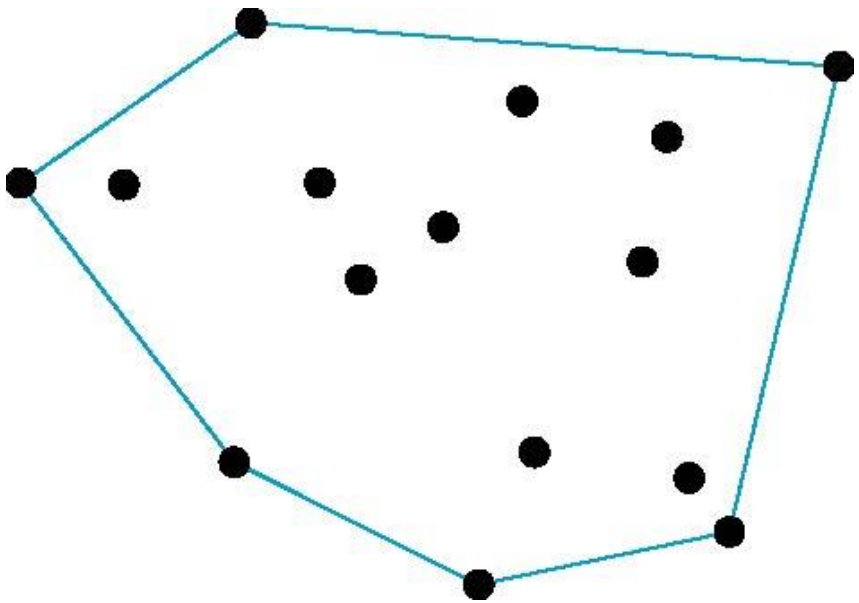
Convexity

- An object is *convex* iff for any two points in the object all points on the line segment between these points are also in the object



Convex Hull

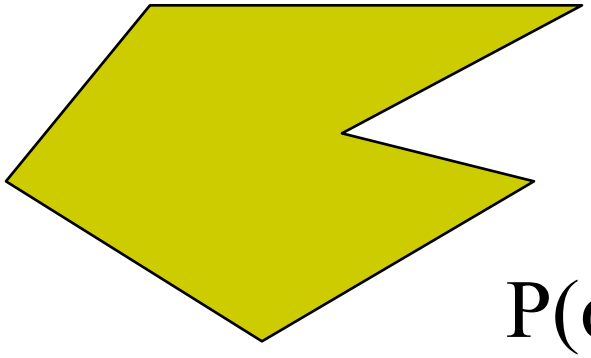
- Smallest convex object containing $P_1, P_2, \ldots\ldots P_n$
- Formed by "shrink wrapping" points



Curves and surfaces

- Curves are one parameter entities of the form $P(\alpha)$ where the function is nonlinear
- Surfaces are formed from two-parameter functions $P(\alpha, \beta)$
  - Linear functions give planes and polygons

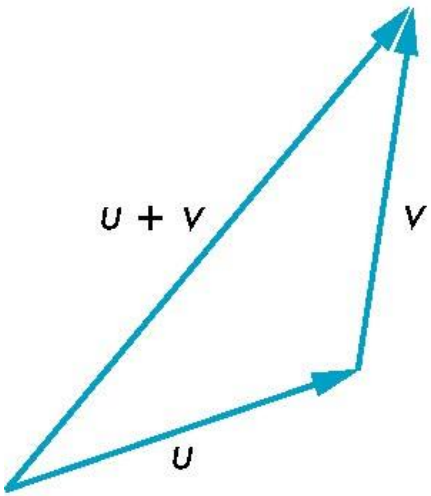
$P(\alpha)$



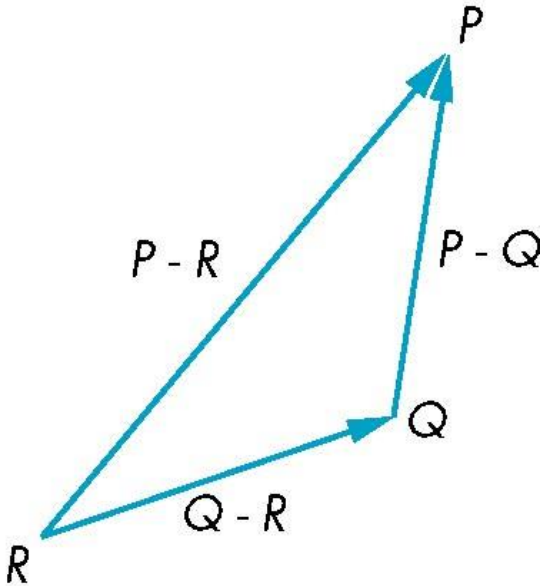
$P(\alpha, \beta)$

Planes

- A plane be determined by a point and two vectors or by three points

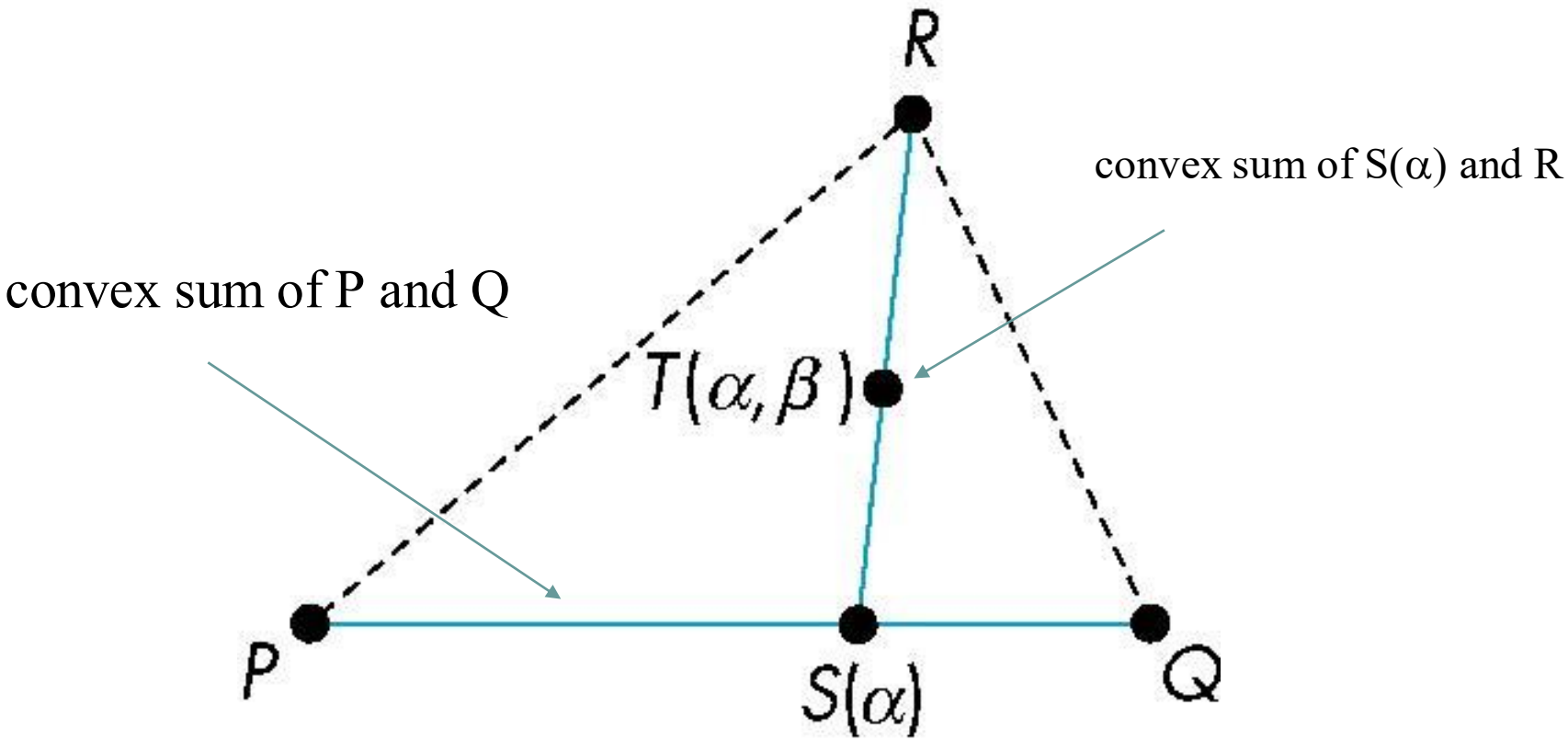


$P(\alpha,\beta)=R+\alpha u+\beta v$



$P(\alpha,\beta)=R+\alpha(Q-R)+\beta(P-Q)$

Triangles

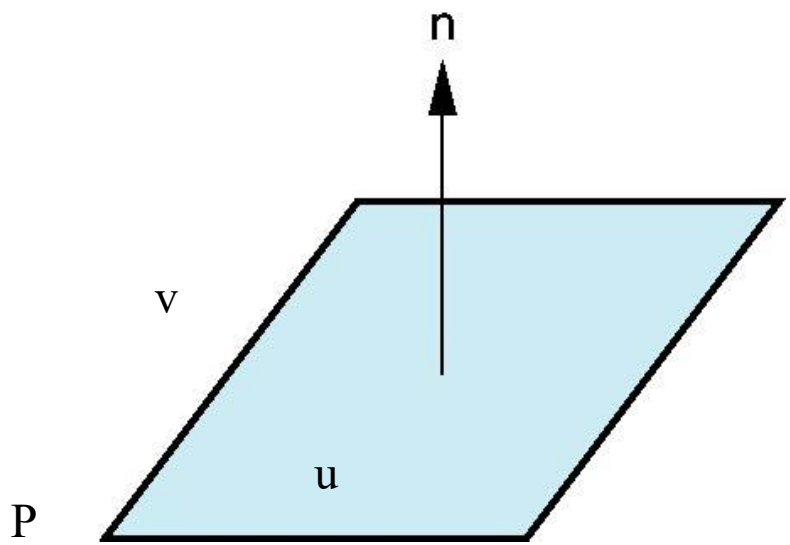


for 0<=α,β<=1, we get all points in triangle

Normals

- Every plane has a vector n normal (perpendicular, orthogonal) to it
- From point-two vector form $P(\alpha,\beta)=R+\alpha u+\beta v$, we know we can use the cross product to find $n = u \times v$ and the equivalent form

$$(P(\alpha)-P) \times n=0$$

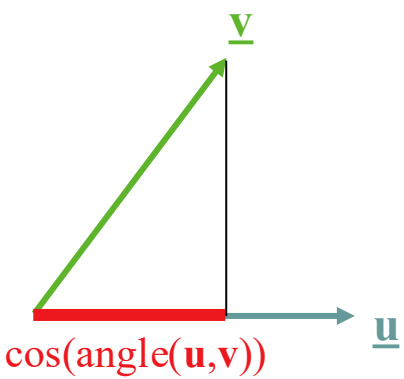


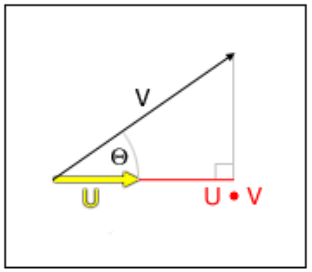
Dot Product

U•V denotes the **dot product** (also called the inner product)

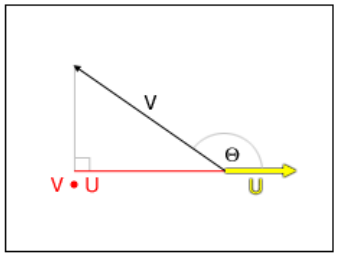
U•V = ||U||·||V||·cos(angle(U,V))

- **U•V** is a scalar.
- **U•V**==0 ⇒ (**U==0** or **V==0** or (**U** and **V** are orthogonal))





V•U = **U•V** > 0 here



V•U = **U•V** < 0 here

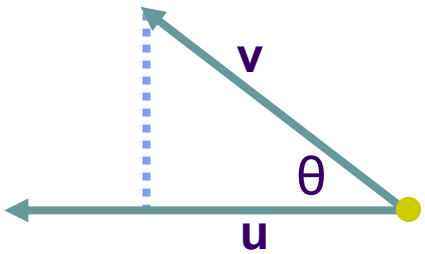
Dot Product

- The *dot product* or, more generally, *inner product* of two vectors is a scalar:

$$\mathbf{v}_1 \bullet \mathbf{v}_2 = \mathbf{x}_1\mathbf{x}_2 + \mathbf{y}_1\mathbf{y}_2 + \mathbf{z}_1\mathbf{z}_2 \qquad \text{(in 3D)}$$

- Useful for many purposes
  - Computing the length of a vector: *length*($\mathbf{v}$) = sqrt($\mathbf{v} \bullet \mathbf{v}$)
  - *Normalizing* a vector, making it unit-length
  - Computing the angle between two vectors:

    $\mathbf{u} \bullet \mathbf{v} = |\mathbf{u}|\,|\mathbf{v}| \cos(\theta)$
  - Checking two vectors for orthogonality
  - *Projecting* one vector onto another

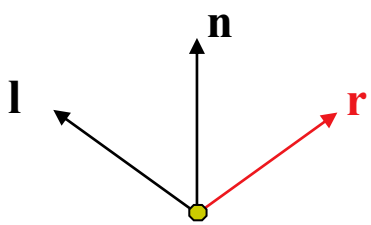


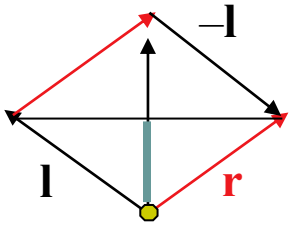
Computing the reflection vector

- Given the unit normal **n** to a mirror surface and the unit direction **l** towards the light, compute the direction **r** of the reflected light.

By symmetry, **l**+**r** is parallel to **n** and has a norm that is twice the length of the projection **n•l** of **l** upon **n**. Hence:

$$\mathbf{r} = 2(\mathbf{n}\bullet\mathbf{l})\mathbf{n}-\mathbf{l}$$





Computing a "shadow" vector (projection)

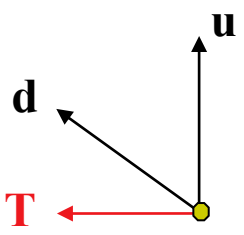
- Given the unit up-vector **u** and the unit direction **d**, compute the shadow **T** of **d** onto the floor (orthogonal to **u**).

T and **u** are orthogonal.

d is the vector sum **T**+ (**d**•**u**)**u** .

Hence:

$$\mathbf{T} = \mathbf{d} - (\mathbf{d} \bullet \mathbf{u})\mathbf{u}$$

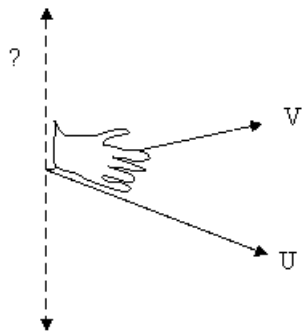


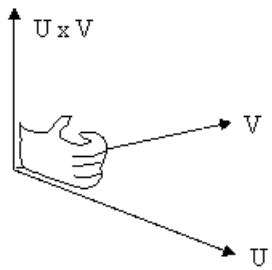
Cross product

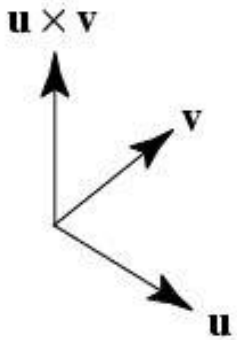
U×**V** denotes the cross product

U×**V** is either **0** or a vector orthogonal to both **U** and **V**

- When **U==0** or **V==0** or **U//V** (parallel) then **U**×**V= 0**
- Otherwise, U×V is orthogonal to both U and V







Cross Product

- The *cross product* or *vector product* of two vectors is a vector:

$$\mathbf{v}_1 \times \mathbf{v}_2 = \begin{bmatrix} y_1 z_2 - y_2 z_1 \\ -(x_1 z_2 - x_2 z_1) \\ x_1 y_2 - x_2 y_1 \end{bmatrix}$$

- The cross product of two vectors is orthogonal to both
- Right-hand rule dictates direction of cross product

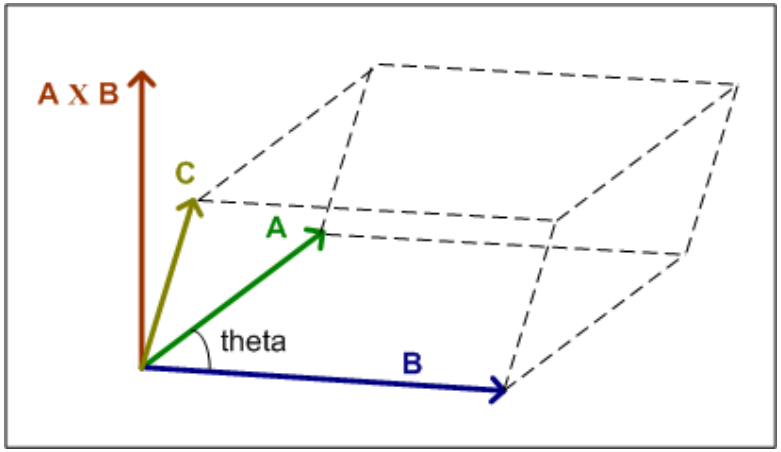
Ex. two edges parallel in 3D?

- Edge(**a**,**b**) is parallel to Edge(**c**,**d**) if **ab**×**cd** == **0**

Mixed product

U•(V×W) is called a **mixed product**

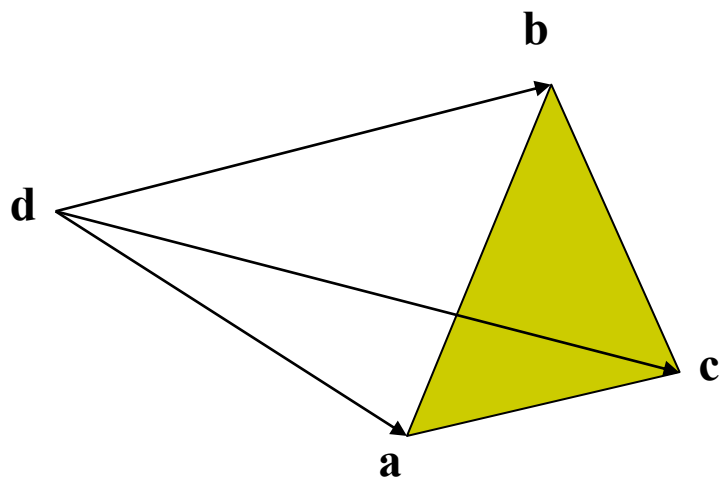
- **U•(V×W)** is a scalar
- **U•(V×W)** ==0 when one of the vectors is null or all 3 are coplanar
- **U•(V×W)** is the determinant | **U V W** |
- **U•(V×W)** ≡ V•(W×U) ≡ – U•(W×V) # cyclic permutation



Testing whether a triangle is front-facing

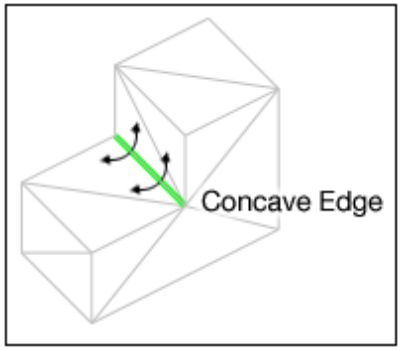
- When does the triangle **a, b, c** appear clockwise from **d**?

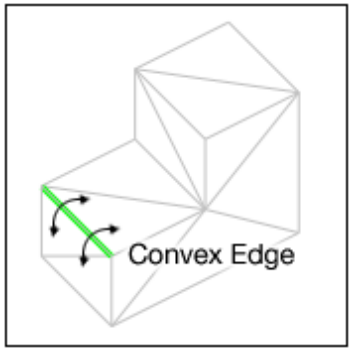
when $\mathbf{da} \bullet (\mathbf{db} \times \mathbf{dc}) > 0$

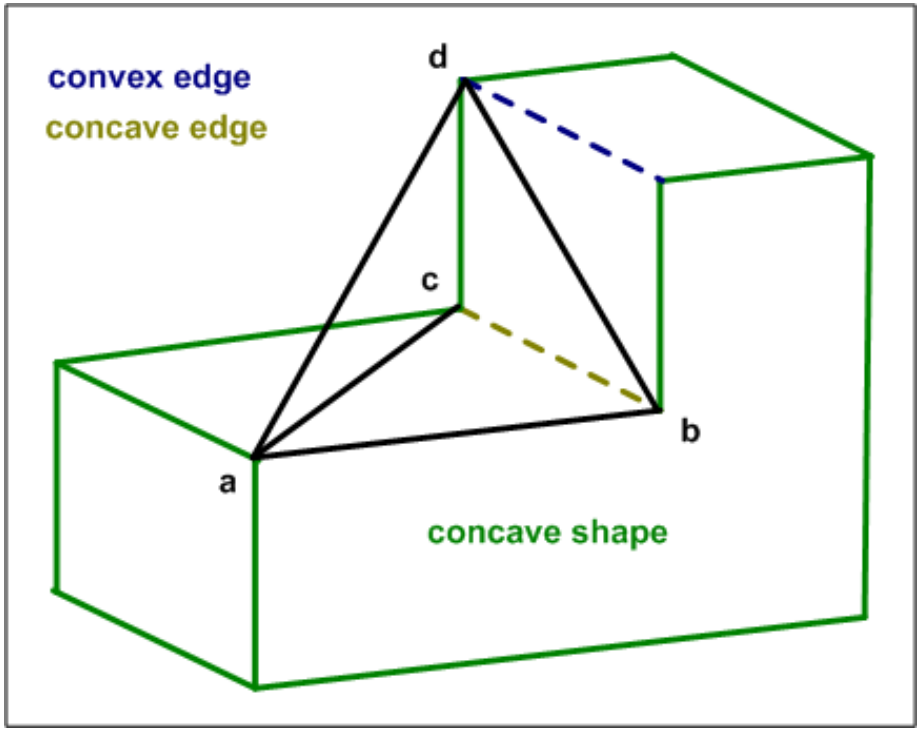


Testing whether an edge is concave

- How to test whether the edge (**c,b**) shared by triangles (**a,b,c**) and (**c,b,d**) is concave?



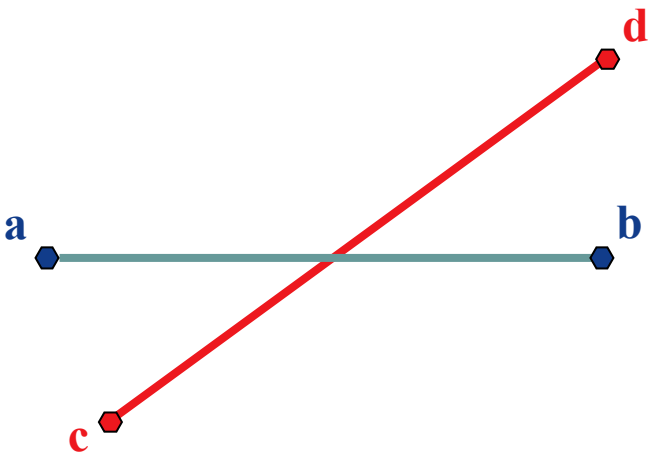


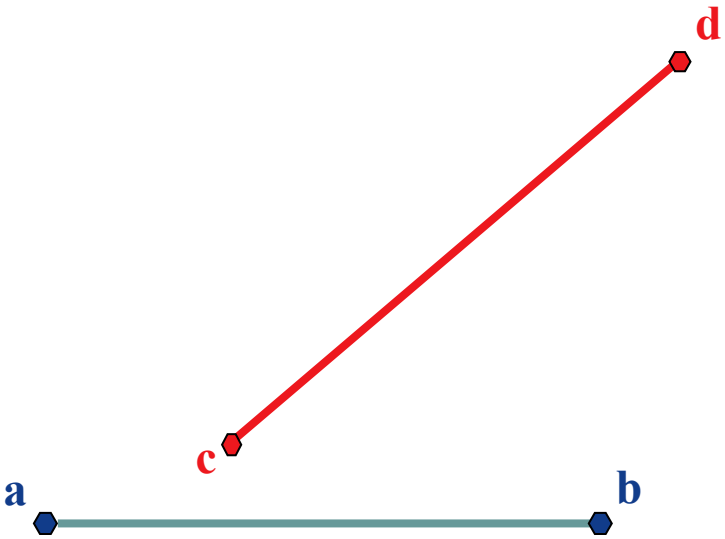


When do two edges intersect in 2D?

- Write a geometric expression that returns *true* when two coplanar edges (**a,b**) and (**c,d**) intersect

$0 \geq (\mathbf{ab} \times \mathbf{ac}) \cdot (\mathbf{ab} \times \mathbf{ad})$ and $0 \geq (\mathbf{cd} \times \mathbf{ca}) \cdot (\mathbf{cd} \times \mathbf{cb})$





When do two edges intersect in 3D?

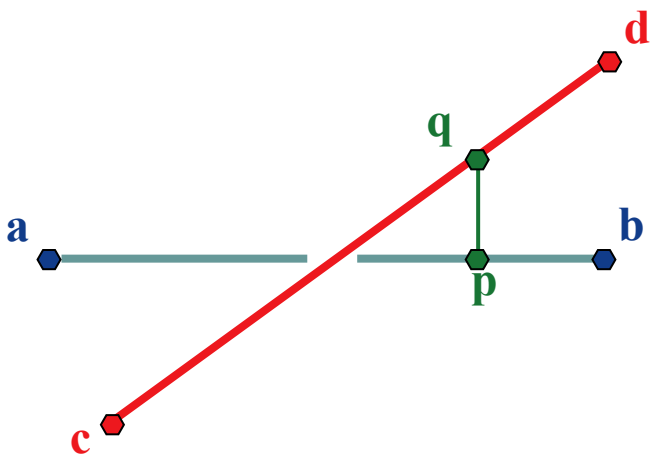
Consider two edges: edge(a,b) and edge(c,d)

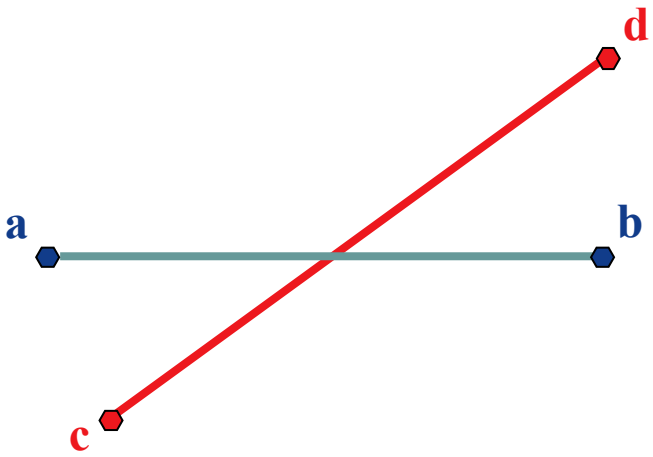
- When are they co-planar? When $z(\mathbf{a},\mathbf{b},\mathbf{c},\mathbf{d})$

- When do they intersect?

  When they are co-planar and

  $0 \geq (\mathbf{ab}\times\mathbf{ac})\bullet(\mathbf{ab}\times\mathbf{ad})$ and $0 \geq (\mathbf{cd}\times\mathbf{ca})\bullet(\mathbf{cd}\times\mathbf{cb})$

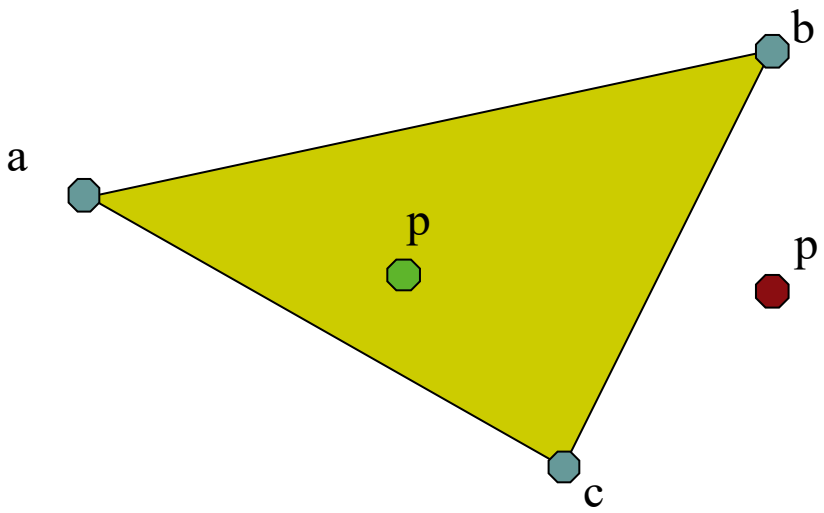




When is a 3D point inside a triangle

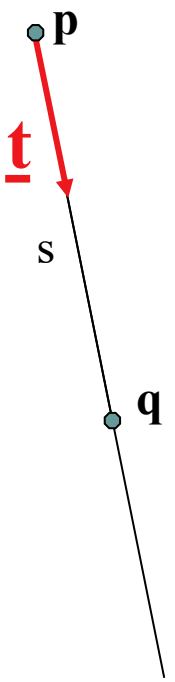
When does point p lie inside triangle with vertices a, b, c ?

When $z(\mathrm{p,a,b,c})$ and $(\mathbf{ab}\times\mathbf{ap})\bullet(\mathbf{bc}\times\mathbf{bp})\geq 0$ and $(\mathbf{bc}\times\mathbf{bp})\bullet(\mathbf{ca}\times\mathbf{cp})\geq 0$



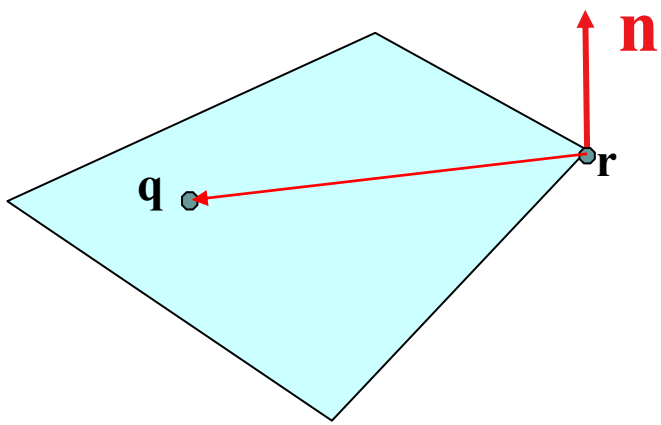
Parametric representation of a line

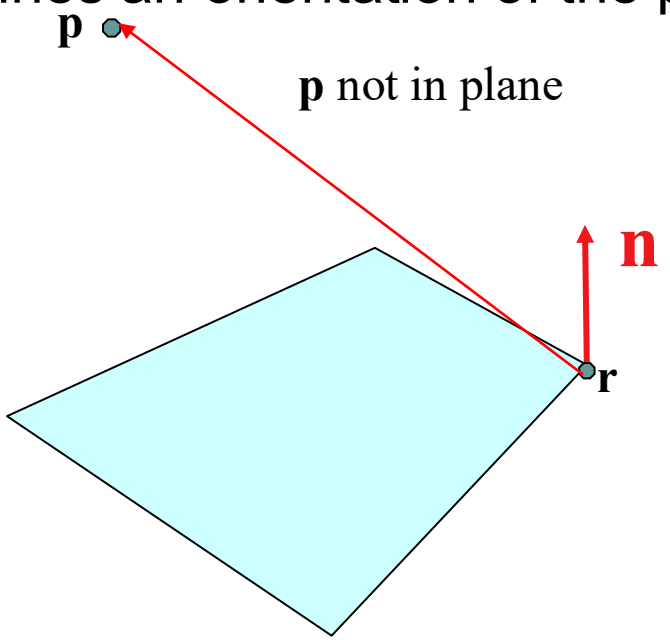
- Let Line(**p**,**t**) be the line through point **p** with tangent **t**
- Its parametric form associates a scalar s with a point **q**(s) on the line
  - s defines the distance from **p** to **q**(s)
  - **q**(s) = **p** + s**t**
  - Note that the direction of **t** gives an orientation to the line (direction where s is positive)
- We can represent a line by **p** and **t**



Implicit representation of a plane

- Let Plane(**r**,**n**) be the plane through point **r** with normal **n**
- Its implicit form states that a point **q** lies on Plane(**r**,**n**) when **rq•n**=0
  - Remember that **rq** = **q** – **r**
- Note that the direction of n defines an orientation of the plane





Line/plane intersection

Compute the point **q** of intersection between Line(**p**,**t**) and Plane(**r**,**n**)

Replacing **q** by **p**+s**t** in **(q**–**r)•n**=0 yields **(p**–**r**+s**t)•n**=0

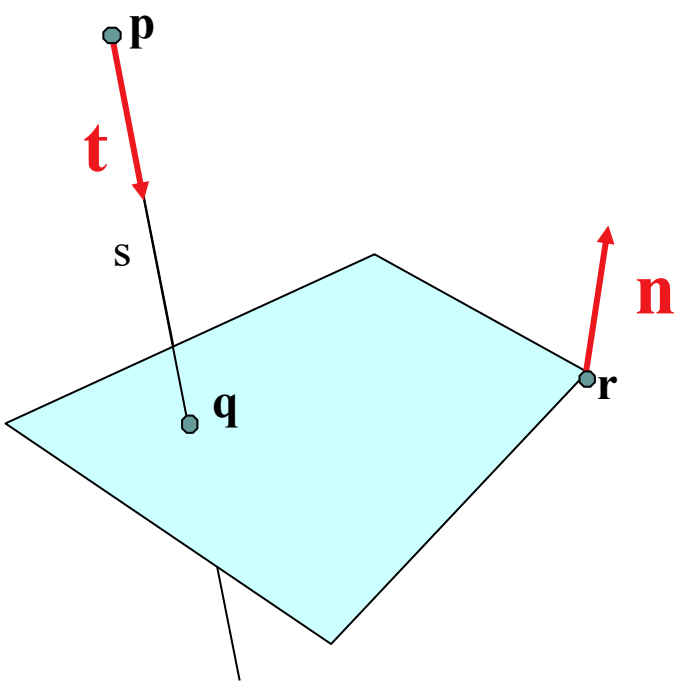
Solving for s yields

rp•n+s**t•n**=0 and

s = – **rp•n / t•n**

s = **pr•n / t•n**

Hence **q** = **p** + (**pr•n)t / (t•n)**



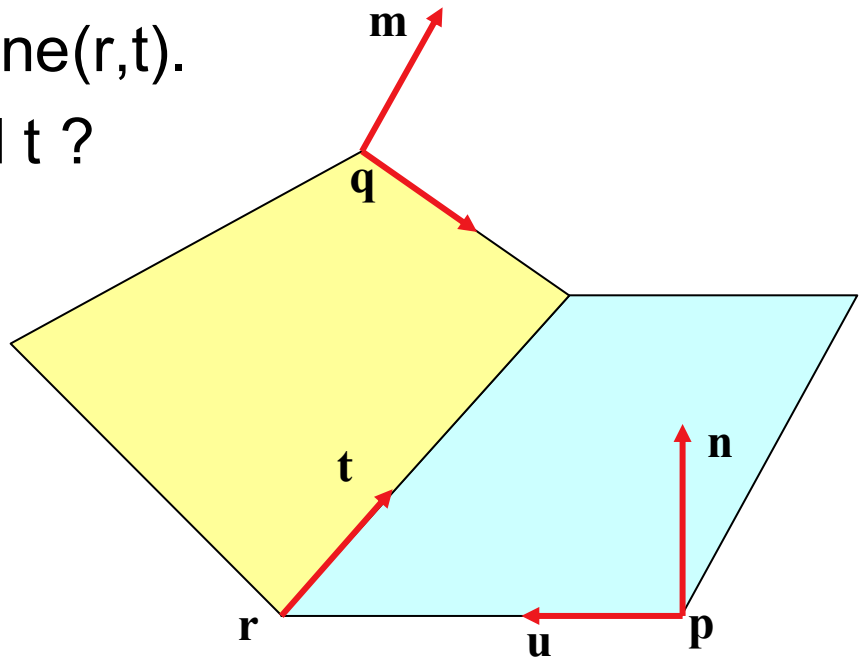
What is the intersection of two planes

- Consider two planes Plane(**p**,**n**) and Plane(**q**,**m**)
- Assume that **n** and **m** are not parallel (i.e. $\mathbf{n}\times\mathbf{m} \neq 0$)
- Their intersection is a line Line(r,t).
- How can one compute r and t ?

$\mathbf{t} := (\mathbf{n}\times\mathbf{m})_u$

let $\mathbf{u} := \mathbf{n}\times\mathbf{t}$

$\mathbf{r} := \mathbf{p}+(\mathbf{pq}\bullet\mathbf{m})\mathbf{u}/(\mathbf{u}\bullet\mathbf{m})$



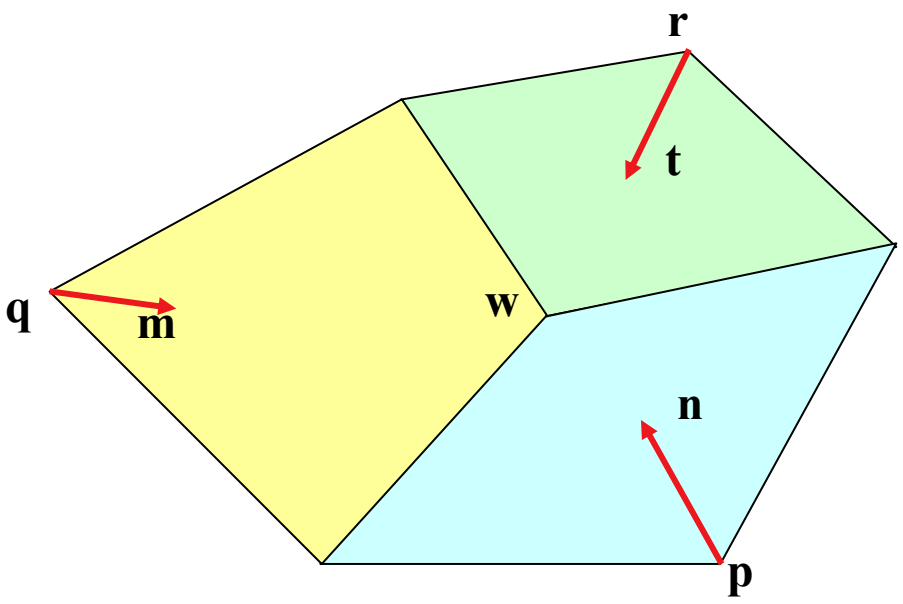
If correct, then provide the derivation.

Otherwise, provided the correct answer.

What is the intersection of three planes

- Consider planes Plane(**p,m**), Plane(**q,n**), and Plane(**r,m**).
- How do you compute their intersection **w** ?

Write **w**=**p**+a**n**+b**m**+c**t** then solve the linear system {**pw**•**n**=0, **qw**•**m**=0, **rw**•**t**=0} for a, b, and c



or compute the line of intersection between two of these planes and then intersect it with the third one.

Implementation

- Points and vectors are each represented by their 3 coordinates
  - $p=(p_x,p_y,p_z)$ and $v=(v_x,v_y,v_z)$
- $U \bullet V = U_x V_x + U_y V_y + U_z V_z$
- $U \times V = (U_y V_z - U_z V_y) - (U_x V_z - U_z V_x) + (U_x V_y - U_y V_x)$
- Implement:
  - Points and vectors
  - Dot, cross, and mixed products
  - z, s, and v functions from mixed products
  - Edge/triangle intersection
  - Lines and Planes
  - Line/Plane, Plane/Plane, and Plane/Plane/Plane intersections
    - Return exception for singular cases (parallelism…)

Linear transformations

- A *linear transformation*:
  - Maps one vector to another
  - Preserves linear combinations
- Thus behavior of linear transformation is completely determined by what it does to a basis
- Turns out any linear transform can be represented by a *matrix*

Matrices

- By convention, matrix element $\mathbf{M}_{rc}$ is located at row *r* and column *c*:

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} & \cdots & \mathbf{M}_{1n} \\ \mathbf{M}_{21} & \mathbf{M}_{22} & \cdots & \mathbf{M}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{M}_{m1} & \mathbf{M}_{m2} & \cdots & \mathbf{M}_{mn} \end{bmatrix}$$

- By (OpenGL) convention, vectors are columns:

$$\mathbf{v} = \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_3 \end{bmatrix}$$

Matrices

- Matrix-vector multiplication applies a linear transformation to a vector:

$$\mathbf{M} \bullet \mathbf{v} = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} & \mathbf{M}_{13} \\ \mathbf{M}_{21} & \mathbf{M}_{22} & \mathbf{M}_{23} \\ \mathbf{M}_{31} & \mathbf{M}_{32} & \mathbf{M}_{33} \end{bmatrix} \begin{bmatrix} \mathbf{v}_x \\ \mathbf{v}_y \\ \mathbf{v}_z \end{bmatrix}$$

- Recall how to do matrix multiplication

Matrix transformations

- A *sequence* or *composition* of linear transformations corresponds to the product of the corresponding matrices
  - Note: the matrices to the *right* affect vector first
  - Note: order of matrices matters!
- The *identity matrix* $\mathbf{I}$ has no effect in multiplication
- Some (not all) matrices have an inverse:

$$\boxed{\mathbf{M}^{-1}(\mathbf{M}(\mathbf{v})) = \mathbf{v}}$$

Homework

- Revise the lectures: L0 and L1
- Practice some simple computation
  - Normal vector of a point
  - Dot and cross product
  - Angle between two vectors (in 3D)
- Write a project proposal
  - Title of your project
  - Describe in general (techniques and solutions)